Some Details

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1 Normalization of Polarization Vectors

First, we give the definition of polarization vectors which is written in terms of adjoint representation

$$\varepsilon_a^\mu = \frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p^J\rangle(\sigma_a)^I{}_J}{4(p\cdot r)} \tag{1}$$

here a runs from 1 to 3. We can use antisymmetric tensor to rewrite this equation,like

$$\varepsilon_a^\mu = \frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p_K\rangle\epsilon^{JK}(\sigma_a)^I{}_J}{4(p\cdot r)} \tag{2}$$

here we choose the left-multiplication convention.

Then, a 2×2 symmetric matrix can be defined like

$$(\sigma_a)^{IJ} = \epsilon^{JK}(\sigma_a)^I{}_K. \tag{3}$$

This means that the original polarization vector can be understood as

$$\begin{aligned}\varepsilon_a^\mu &= -\frac{\langle p_I|(r\cdot\Gamma)\Gamma^\mu|p_J\rangle}{4(p\cdot r)}(\sigma_a)^{IJ} \\ &= -\varepsilon_{IJ}^\mu(\sigma_a)^{IJ}\end{aligned} \tag{4}$$

here I, J are fundamental little group indices, runs from 1 to 2.

And we know that the reasonable polarization vector for physical state should be symmetric with little group indices, it means that we need to symmetrize the indices I and J as $(IJ) = \frac{1}{2}(IJ + JI)$. Because of the symmetry of $(\sigma_a)^{IJ}$, so it can be obtained that

$$\varepsilon_a^\mu = -(\varepsilon_{(IJ)}^\mu + \varepsilon_{[IJ]}^\mu)(\sigma_a)^{IJ} = -\varepsilon_{(IJ)}^\mu(\sigma_a)^{IJ} \tag{5}$$

which means we only need to consider the symmetric part of ε_{IJ}^μ.

Now, we want to compute the normalization for ε_{IJ}^μ, and after multiplying the symmetric matrix, it should go back to the normalization

$$\varepsilon_a^\mu\varepsilon_{b,\mu} = -\delta_{ab} \tag{6}$$

The method to compute the normalization is just multiplying two polarization vectors directly, like

$$\varepsilon^{\mu}_{(IJ)}\varepsilon_{(KL),\mu} = \frac{\langle p_I|(r\cdot\Gamma)\Gamma^{\mu}|p_J\rangle}{4(p\cdot r)} \times \frac{\langle p_K|(r\cdot\Gamma)\Gamma_{\mu}|p_L\rangle}{4(p\cdot r)} \tag{7}$$

here we need to use a special 5d equality

$$(\Gamma^{\mu})_A^{\ B}(\Gamma_{\mu})_C^{\ D} = -2\Omega_{AC}\Omega^{BD} + 2\delta_A^{\ D}\delta_C^{\ B} - \delta_A^{\ B}\delta_C^{\ D} \tag{8}$$

and the dirac operator can be decomposed to

$$\not{p} = p\cdot\Gamma = p_A^{\ B} = |p_I\rangle_A\langle p^I|^B \tag{9}$$

which help us to rewrite the equation (7) to be

$$\begin{aligned}\varepsilon^{\mu}_{IJ}\varepsilon_{KL,\mu} &= \frac{\langle p_I|^A|r_M\rangle_A\langle r^M|^B(\Gamma^{\mu})_B^{\ C}|p_J\rangle_C\langle p_K|^D|r_N\rangle_D\langle r^N|^E(\Gamma_{\mu})_D^{\ F}|p_L\rangle_F}{16(p\cdot r)^2}\\ &= \frac{\langle p_I|r_M\rangle\langle p_K|r_N\rangle}{16(p\cdot r)^2}\times\langle r^M|^B(\Gamma^{\mu})_B^{\ C}|p_J\rangle_C\langle r^N|^E(\Gamma_{\mu})_E^{\ F}|p_L\rangle_F\end{aligned} \tag{10}$$

After plugging the equation (8) into the second part of (10), it becomes

$$\begin{aligned}&\langle r^M|^B|p_J\rangle_C\langle r^N|^E|p_L\rangle_F(-2\Omega_{BE}\Omega^{CF} + 2\delta_B^{\ F}\delta_E^{\ C} - \delta_B^{\ C}\delta_E^{\ F})\\ &= -2\langle r^M|r^N\rangle\langle p_L|p_J\rangle + 2\langle r^M|p_L\rangle\langle r^N|p_J\rangle - \langle r^M|p_J\rangle\langle r^N|p_L\rangle\end{aligned} \tag{11}$$

The first term of (11) equals to 0 because the equality

$$\langle p_I|p_J\rangle = 0 \tag{12}$$

and the remaining two term can be contracted with the first part of (10), which helps to obtain the following two terms

$$2\langle p_I|r_M\rangle\langle r^M|p_L\rangle\langle p_K|r_N\rangle\langle r^N|p_J\rangle - \langle p_I|r_M\rangle\langle r^M|p_J\rangle\langle p_K|r_N\rangle\langle r^N|p_L\rangle \tag{13}$$

It is known that

$$\langle p_I|(r\cdot\Gamma)|p_J\rangle = \mathrm{Tr}[|p_J\rangle\langle p_I|(r\cdot\Gamma)] = \frac{1}{2}\epsilon_{IJ}\mathrm{Tr}[(p\cdot\Gamma)(r\cdot\Gamma)] = 2\epsilon_{IJ}(p\cdot r) \tag{14}$$

so the formal equation can be rewritten like

$$8(p\cdot r)^2\epsilon_{IL}\epsilon_{KJ} - 4(p\cdot r)^2\epsilon_{IJ}\epsilon_{KL} \tag{15}$$

Finally, we need to symmetrize the indices (IJ) and (KL), so that we obtain

$$\varepsilon^{\mu}_{IJ}\varepsilon_{KL,\mu} = \frac{1}{8(p\cdot r)^2}(p\cdot r)^2(\epsilon_{IL}\epsilon_{KJ} + \epsilon_{JL}\epsilon_{KI} + \epsilon_{IK}\epsilon_{LJ} + \epsilon_{JK}\epsilon_{LI}) - \frac{1}{16(p\cdot r)^2}(p\cdot r)^2(\epsilon_{(IJ)}\epsilon_{(KL)}) \tag{16}$$

because ϵ itself is an antisymmetric tensor so it will become zero under the symmetrization, so that the second term is vanishing.

After some rearrangement of indices, we can obtain

$$\varepsilon^{\mu}_{IJ}\varepsilon_{KL,\mu} = \frac{1}{4}(\epsilon_{IL}\epsilon_{KJ} + \epsilon_{IK}\epsilon_{LJ}) \tag{17}$$

Unfortunately, when we want to reproduce the original normalization relation, we will obtain something like

$$\begin{aligned}
\varepsilon^{\mu}_{a}\varepsilon_{b,\mu} &= (-\varepsilon^{\mu}_{(IJ)}(\sigma_a)^{IJ})(-\varepsilon^{\mu}_{(KL)}(\sigma_b)^{KL}) \\
&= \frac{1}{4}(\epsilon_{IL}\epsilon_{KJ} + \epsilon_{IK}\epsilon_{LJ})(\sigma_a)^{IJ}(\sigma_b)^{KL} \\
&= \frac{1}{4}\epsilon_{IL}\epsilon_{KJ}\epsilon^{JM}(\sigma_a)^{I}{}_{M}\epsilon^{LN}(\sigma_b)^{K}_{\ N} + \frac{1}{4}\epsilon_{IK}\epsilon_{LJ}\epsilon^{IM}(\sigma_a)^{J}{}_{M}\epsilon^{LN}(\sigma_b)^{K}_{\ N} \\
&= \frac{1}{4}\delta_{I}^{\ N}\delta_{K}^{\ M}(\sigma_a)^{I}{}_{M}(\sigma_b)^{K}_{\ N} + \frac{1}{4}\delta_{K}^{\ M}\delta_{J}^{\ N}(\sigma_a)^{J}{}_{M}(\sigma_b)^{K}_{\ N} \\
&= \frac{1}{4}2\mathrm{tr}(\sigma_a\sigma_b) \\
&= \delta_{ab}
\end{aligned} \tag{18}$$

2 Completeness Relation of Polarization Vectors

For this purpose, we need to compute the following formula

$$\varepsilon^{\mu}_{a}\varepsilon^{a,\nu} = ? \tag{19}$$

If we multiply the momentum p_μ, it should become vanishing so that the completeness relation should look like

$$\varepsilon^{\mu}_{a}\varepsilon^{a,\nu} = k(-\eta^{\mu\nu} + \frac{p^\mu r^\nu + p^\nu r^\mu}{(p\cdot r)^2}) \tag{20}$$

here k is just a overall coefficient.

Here, we choose to compute the completeness relation for polarization vectors with fundamental indices

$$\varepsilon^{\mu}_{IJ}\varepsilon^{IJ,\nu} = a(-\eta^{\mu\nu} + \frac{p^\mu r^\nu + p^\nu r^\mu}{(p\cdot r)^2}) \tag{21}$$

with a different coefficient.

For symplisity, we start with the convention from paper arxiv:2202.08257, in which the polarization vector looks like

$$\epsilon^{\mu}_{IJ} = \frac{\langle p_{(I}|\Gamma^\mu|r_{J)}\rangle}{\sqrt{2}} \tag{22}$$

with the normalization

$$\langle p^I|r^J\rangle = \epsilon^{IJ} \tag{23}$$

which means the following condition

$$2(p \cdot r) = 1 \tag{24}$$

The method to compute this completeness relation is still just multiplying them together like

$$\varepsilon^{\mu}_{(IJ)}\varepsilon^{(IJ),\nu} = \frac{\langle p_{(I}|\Gamma^{\mu}|r_{J)}\rangle}{\sqrt{2}} \times \frac{\langle p^{(I}|\Gamma^{\nu}|r^{J)}\rangle}{\sqrt{2}} \tag{25}$$

It can be decomposed into 4 pices IJ^{IJ}, IJ^{JI}, JI^{IJ}, JI^{JI}, and the first one can be computed like

$$\begin{aligned}\langle p_I|\Gamma^{\mu}|r_J\rangle \times \langle p^I|\Gamma^{\nu}|r^J\rangle &= -\langle p_I|\Gamma^{\mu}|r_J\rangle\langle r^J|\Gamma^{\nu}|p^I\rangle \\ &= -\mathrm{Tr}[|p^I\rangle\langle p_I|\Gamma^{\mu}|r_J\rangle\langle r|^J\Gamma^{\nu}] \\ &= \mathrm{Tr}[(p\cdot\Gamma)\Gamma^{\mu}(r\cdot\Gamma)\Gamma^{\nu}] \\ &= 4(p^{\mu}r^{\nu} + p^{\nu}r^{\mu} - (p\cdot r)\eta^{\mu\nu})\end{aligned} \tag{26}$$

where we use a nontrival fact that

$$\langle p^I|\Gamma^{\nu}|r^J\rangle = -\langle r^J|\Gamma^{\nu}|p^I\rangle \quad (\text{obtained from} \quad C^T\Gamma^{\mu}C = -(\Gamma^{\mu})^T) \tag{27}$$

The fourth piece is nearly the same as the first one

$$\langle p_J|\Gamma^{\mu}|r_I\rangle \times \langle p^J|\Gamma^{\nu}|r^I\rangle = 4(p^{\mu}r^{\nu} + p^{\nu}r^{\mu} - (p\cdot r)\eta^{\mu\nu}) \tag{28}$$

The remaining two need some extra efforts, here I compute it explictly

$$\langle p_I|\Gamma^{\mu}|r_J\rangle \times \langle p^J|\Gamma^{\nu}|r^I\rangle \tag{29}$$

In terms of two massless spinors that satisfy (23), the $USp(2,2)$ identity operator can be written as

$$|r_a\rangle_A\langle p^a|^B + |p_a\rangle_A\langle r^a|^B = \delta_A^{\ B} \tag{30}$$

It can be used to rewrite the equation (29) like

$$\begin{aligned}&\langle p_I|^A(\Gamma^{\mu})_A^{\ B}|r_J\rangle_B\langle p^J|^C(\Gamma^{\nu})_C^{\ D}|r^I\rangle_D \\ &= \langle p_I|^A(\Gamma^{\mu})_A^{\ B}(\delta_B^{\ C} - |p_J\rangle_B\langle r^J|^C)(\Gamma^{\nu})_C^{\ D}|r^I\rangle_D \\ &= \langle p_I|^A(\Gamma^{\mu})_A^{\ B}(\Gamma^{\nu})_B^{\ D}|r^I\rangle_D - \langle p_I|\Gamma^{\mu}|p_J\rangle\langle r^J|\Gamma^{\nu}|r^I\rangle\end{aligned} \tag{31}$$

And similarly for

$$\begin{aligned}\langle p_J|\Gamma^{\mu}|r_I\rangle \times \langle p^I|\Gamma^{\nu}|r^J\rangle &= \langle p^I|\Gamma^{\nu}|r^J\rangle \times \langle p_J|\Gamma^{\mu}|r_I\rangle \\ &= \langle p^I|^A(\Gamma^{\nu})_A^{\ B}(-|p^J\rangle_B\langle r_J|^{\ C} - \delta_B^{\ C})(\Gamma^{\mu})_C^{\ D}|r_I\rangle_D \\ &= -\langle p^I|^A(\Gamma^{\nu})_A^{\ B}(\Gamma^{\mu})_B^{\ D}|r_I\rangle_D - \langle p^I|\Gamma^{\nu}|p^J\rangle\langle r_J|\Gamma^{\mu}|r_I\rangle\end{aligned} \tag{32}$$

Now we can combine these two terms

$$\begin{aligned}&\langle p_I|^A(\Gamma^{\mu})_A^{\ B}(\Gamma^{\nu})_B^{\ D}|r^I\rangle_D - \langle p_I|\Gamma^{\mu}|p_J\rangle\langle r^J|\Gamma^{\nu}|r^I\rangle - \langle p^I|^A(\Gamma^{\nu})_A^{\ B}(\Gamma^{\mu})_B^{\ D}|r_I\rangle_D - \langle p^I|\Gamma^{\mu}|p^J\rangle\langle r_J|\Gamma^{\nu}|r_I\rangle \\ &= \langle p_I|(\Gamma^{\mu}\Gamma^{\nu} + \Gamma^{\nu}\Gamma^{\mu})|r^I\rangle - \langle p_I|\Gamma^{\mu}|p_J\rangle\langle r^J|\Gamma^{\nu}|r^I\rangle - \langle p^I|\Gamma^{\nu}|p^J\rangle\langle r_J|\Gamma^{\mu}|r_I\rangle \\ &= 2\eta^{\mu\nu}\langle p_I|r^I\rangle - (2p^{\mu}\epsilon_{IJ})(-2r^{\nu}\epsilon^{JI}) - (-2p^{\nu}\epsilon^{IJ})(2r^{\mu}\epsilon_{JI}) \\ &= 4\eta^{\mu\nu} + 8p^{\mu}r^{\nu} + 8p^{\nu}r^{\mu}\end{aligned} \tag{33}$$

Above all, we can conclude that

$$\begin{aligned}\varepsilon^{\mu}_{(IJ)}\varepsilon^{(IJ),\nu} &= \frac{1}{2}\times\frac{1}{8}[8(p^{\mu}r^{\nu}+p^{\nu}r^{\mu}-(p\cdot r)\eta^{\mu\nu})+4\eta^{\mu\nu}+8p^{\mu}r^{\nu}+8p^{\nu}r^{\mu}] \\ &\overset{?}{=} 2p^{(\mu}r^{\nu)} \end{aligned} \tag{34}$$

3 Completeness Relation Review

Under the following definition of polarization vector

$$\varepsilon^{\mu}_{a} = \frac{\langle p_I|(r\cdot\Gamma)\Gamma^{\mu}|p_J\rangle}{4(p\cdot r)}(\sigma_a)^{I}{}_{K}\epsilon^{KJ} \tag{35}$$

and same for

$$\varepsilon^{\mu}_{a} = \frac{\langle p^I|(r\cdot\Gamma)\Gamma^{\mu}|p^J\rangle}{4(p\cdot r)}(\sigma_a)^{K}{}_{J}\epsilon_{KI}, \tag{36}$$

the polarization vector living in the adjoint representation can be defined as

$$\varepsilon^{\mu}_{IJ}(p;r) := \frac{\langle p_{(I}|(r\cdot\Gamma)\Gamma^{\mu}|p_{J)}\rangle}{4(p\cdot r)} \tag{37}$$

$$\varepsilon^{IJ,\mu}(p;r) := \frac{\langle p^{(I}|(r\cdot\Gamma)\Gamma^{\mu}|p^{J)}\rangle}{4(p\cdot r)}. \tag{38}$$

The corresponding complex conjugate can be obtained

$$(\varepsilon^{\mu}_{IJ}(p;r))^{*} = -\frac{\langle p^{(I}|\Gamma^{\mu}(r\cdot\Gamma)|p^{J)}\rangle}{4(p\cdot r)} \tag{39}$$

and similarly

$$(\varepsilon^{IJ,\mu}(p;r))^{*} = -\frac{\langle p_{(I}|\Gamma^{\mu}(r\cdot\Gamma)|p_{J)}\rangle}{4(p\cdot r)}. \tag{40}$$

For the consistency of index placement with the summation convention, we can define the following quantities

$$\varepsilon^{*\mu}_{IJ}(p;r) := (\varepsilon^{IJ,\mu}(p;r))^{*} = -\frac{\langle p_{(I}|\Gamma^{\mu}(r\cdot\Gamma)|p_{J)}\rangle}{4(p\cdot r)}, \tag{41}$$

$$\varepsilon^{*IJ,\mu}(p;r) := (\varepsilon^{\mu}_{IJ}(p;r))^{*} = -\frac{\langle p^{(I}|\Gamma^{\mu}(r\cdot\Gamma)|p^{J)}\rangle}{4(p\cdot r)} \tag{42}$$

Then the completeness relation can be computed like

$$\varepsilon^{*\,IJ,\mu}\,\varepsilon^{\nu}_{IJ} = -\frac{\langle p^{(I}|\Gamma^{\mu}(r\cdot\Gamma)|p^{J)}\rangle}{4(p\cdot r)}\frac{\langle p_{(I}|(r\cdot\Gamma)\Gamma^{\nu}|p_{J)}\rangle}{4(p\cdot r)} \tag{43}$$

It includes 4 terms and can be grouped into two part, the first part can be computed like

$$\begin{aligned}
&\langle p^I|\Gamma^\mu(r\cdot\Gamma)|p^J\rangle\langle p_I|(r\cdot\Gamma)\Gamma^\nu|p_J\rangle \\
&= -\langle p^I|\Gamma^\mu(r\cdot\Gamma)|p^J\rangle\langle p_J|\Gamma^\nu(r\cdot\Gamma)|p_I\rangle \\
&= -\mathrm{tr}[|p_I\rangle\langle p^I|\Gamma^\mu(r\cdot\Gamma)|p^J\rangle\langle p_J|\Gamma^\nu(r\cdot\Gamma)] \\
&= \mathrm{tr}[(p\cdot\Gamma)\Gamma^\mu(r\cdot\Gamma)(p\cdot\Gamma)\Gamma^\nu(r\cdot\Gamma)]
\end{aligned} \tag{44}$$

Here we need to compute trace for the product of 6 gamma matrices, and I just give the answer

$$\begin{aligned}
\mathrm{Tr}\Big(\Gamma^a\Gamma^b\Gamma^c\Gamma^d\Gamma^e\Gamma^f\Big) = 4\Big(&\eta^{ab}\eta^{cd}\eta^{ef} - \eta^{ab}\eta^{ce}\eta^{df} + \eta^{ab}\eta^{cf}\eta^{de} \\
&- \eta^{ac}\eta^{bd}\eta^{ef} + \eta^{ac}\eta^{be}\eta^{df} - \eta^{ac}\eta^{bf}\eta^{de} \\
&+ \eta^{ad}\eta^{bc}\eta^{ef} - \eta^{ad}\eta^{be}\eta^{cf} + \eta^{ad}\eta^{bf}\eta^{ce} \\
&- \eta^{ae}\eta^{bc}\eta^{df} + \eta^{ae}\eta^{bd}\eta^{cf} - \eta^{ae}\eta^{bf}\eta^{cd} \\
&+ \eta^{af}\eta^{bc}\eta^{de} - \eta^{af}\eta^{bd}\eta^{ce} + \eta^{af}\eta^{be}\eta^{cd}\Big).
\end{aligned} \tag{45}$$

A compact way to write the result is

$$\mathrm{Tr}\Big(\Gamma^{a_1}\cdots\Gamma^{a_6}\Big) = 4\sum_{\text{pairings}}(-1)^\sigma\,\eta^{a_{i_1}a_{j_1}}\eta^{a_{i_2}a_{j_2}}\eta^{a_{i_3}a_{j_3}}.$$

Here the sum runs over all pairwise partitions (pairings) of the index set $\{1,\ldots,6\}$ into three disjoint pairs $\{(i_1,j_1),(i_2,j_2),(i_3,j_3)\}$.Here $(-1)^\sigma$ denotes the sign (parity) of the permutation needed to reorder the sequence so that the members of each pair become adjacent.

Using this formula, we can rewrite the equation (44)

$$\begin{aligned}
(44) = \quad & p^\mu r^\nu(p\cdot r) - p^\mu r^\nu(p\cdot r) + p^\mu p^\nu r^2 \\
&- p^\mu r^\nu(p\cdot r) + \eta^{\mu\nu}(p\cdot r)^2 - p^\nu r^\mu(p\cdot r) \\
&+ r^\mu r^\nu p^2 - \eta^{\mu\nu}p^2r^2 + r^\mu r^\nu p^2 \\
&- p^\nu r^\mu(p\cdot r) + p^\nu p^\mu r^2 - p^\nu r^\mu(p\cdot r) \\
&+ p^\nu r^\mu(p\cdot r) - p^\mu r^\nu(p\cdot r) + \eta^{\mu\nu}(p\cdot r)^2 \\
&= 4(-2p^\mu r^\nu(p\cdot r) - 2p^\nu r^\mu(p\cdot r) + \eta^{\mu\nu}(p\cdot r)^2)
\end{aligned} \tag{46}$$

in which many terms cancel with each other.

Similarly, the second part can be computed like

$$\begin{aligned}
&\langle p^I|\Gamma^\mu(r\cdot\Gamma)|p^J\rangle\langle p_J|(r\cdot\Gamma)\Gamma^\nu|p_I\rangle \\
&= \mathrm{tr}[|p_I\rangle\langle p^I|\Gamma^\mu(r\cdot\Gamma)|p^J\rangle\langle p_J|(r\cdot\Gamma)\Gamma^\nu] \\
&= -\mathrm{tr}[(p\cdot\Gamma)\Gamma^\mu(r\cdot\Gamma)(p\cdot\Gamma)(r\cdot\Gamma)\Gamma^\nu] \\
&= -\mathrm{tr}[(p\cdot\Gamma)\Gamma^\mu(r\cdot\Gamma)(p\cdot\Gamma)\{(r\cdot\Gamma),\Gamma^\nu\}] + \mathrm{tr}[(p\cdot\Gamma)\Gamma^\mu(r\cdot\Gamma)(p\cdot\Gamma)\Gamma^\nu(r\cdot\Gamma)]
\end{aligned} \tag{47}$$

It can be noticed that the first term equals to 0 and the second term is just the same as the result for first part. In short, the two part give us the same result.

Above all. the completeness relation can be written as

$$
\begin{aligned}
\varepsilon^{*\,IJ,\mu}\,\varepsilon^{\nu}_{IJ} &= -\frac{1}{16(p\cdot r)^2}\times\frac{1}{4}\times\Big[4\times 4(-2p^\mu r^\nu(p\cdot r)-2p^\nu r^\mu(p\cdot r)+\eta^{\mu\nu}(p\cdot r)^2)\Big] \\
&= -\frac{1}{2}\left[\frac{-p^\mu r^\nu - p^\nu r^\mu}{(p\cdot r)}+\eta^{\mu\nu}\right] \\
&= -\frac{1}{2}\eta^{\mu\nu}+\frac{p^{(\mu}r^{\nu)}}{(p\cdot r)}
\end{aligned} \tag{48}
$$

We can quickly check its correctness by multiplying a momentum

$$
p_\mu\varepsilon^{*\,IJ,\mu}\,\varepsilon^{\nu}_{IJ} = -\frac{1}{2}p^\nu+\frac{\frac{1}{2}p^2r^\nu+\frac{1}{2}p^\nu(p\cdot r)}{(p\cdot r)} = 0 \tag{49}
$$

4 All for 5 dimension

4.1 Spinor-Helicty in 5D

The Lorentz group in 5D is $\mathrm{Spin}(5)\cong\mathrm{USp}(4)\cong\mathrm{Sp}(4,\mathbb{C})\cap\mathrm{SU}(4)$,we can also consider the non-compact version $\mathrm{SO}(1,4)$, which has an isomorphism $\mathrm{SO}(1,4)\cong\mathrm{USp}(2,2)$. For a 5-momentum vector p^μ, we can identify it with a 4×4 antisymmetric matrix

$$
p_{AB} = p_\mu\gamma^\mu_{AB} \equiv -p_\mu(\gamma^\mu)_A^{\ C}\Omega_{CB} = -(\not{p}\,\Omega)_{AB} \tag{50}
$$

Here we choose a symplectic form to be the metric

$$
\Omega_{AB} = \begin{pmatrix}\epsilon_{\alpha\beta} & 0 \\ 0 & -\epsilon^{\dot\alpha\dot\beta}\end{pmatrix} = \Omega^{BA} \tag{51}
$$

where we use the fact that fundamental indices A, B for $\mathrm{USp}(2,2)$ can be decomposed to $\mathrm{SL}(2,\mathbb{C})$ indices like $A=\alpha\oplus\dot\alpha$, and the normalization for 2 dimension Levi-Civita tensor is $\epsilon^{12}=\epsilon_{21}=1$.

We use metric Ω to lower and raise the indices like

$$
X^A = \Omega^{AB}X_B, \qquad X_A = \Omega_{AB}X^B \tag{52}
$$

where X are spinors belongs to $\mathrm{USp}(2,2)$.

Also, from the defination (50), we know the gamma matrices with lower indices can be defined like

$$
\gamma_{AB} = \Omega_{BC}\gamma_A^{\ C} \tag{53}
$$

similary the gamma matrix with upper indices can be obtained by

$$
\gamma^{AB} = \Omega^{AC}\gamma_C^{\ B}. \tag{54}
$$

The gamma matrices with lower indices are antisymmetric and Ω-traceless

$$
\gamma_{AB} = \gamma_{BA}, \qquad \gamma_{AB}\Omega^{AB} = 0 \tag{55}
$$

The determinant of p_{AB} evaluates to $\mathrm{Det}(p_{AB}) = \frac{1}{4}(p_{AB}p^{AB})^2 = (p^2)^2$. For 5D massless particle, $p^2 = 0 \Rightarrow \mathrm{Det}(p_{AB} = 0)$, so the rank of this matrix equals to 2. Beacuse the little group of 5D massless particle is $\mathrm{SO}(3) \cong \mathrm{SU}(2)$, so the matrix p_{AB} can be decomposed to be the outer product of two 4×2 matrix, here specifically, the $\mathrm{USp}(2,2)$ spinors with $\mathrm{SU}(2)$ little group indices

$$p_{AB} = \lambda_A^{\ a}\lambda_{Ba} = |p^a\rangle_A |p_a\rangle_B, \qquad p_A^{\ B} = \lambda_A^{\ a}\lambda^B_{\ a} = |p^a\rangle_A \langle p_a|^B \tag{56}$$

with a, b runs over $1, 2$. A little group transformation that preserves p_{AB} behaves like $|p^a\rangle_A \to t^a_{\ b}|p^b\rangle_A$, here $t^a_{\ b}$ refers to the $\mathrm{SU}(2)$ little group transformation.

Since the little groups of massive 4D momentum and massless 5D momentum are identified, there exist a consistent embedding of the former into the latter.

$$\lambda_A^{\ a} = \begin{pmatrix} \lambda_\alpha^{\ a} \\ \tilde{\lambda}^{\dot{\alpha} a} \end{pmatrix}, \qquad \Omega_{AB} = \begin{pmatrix} -\epsilon_{\alpha\beta} & 0 \\ 0 & \epsilon^{\dot{\alpha}\dot{\beta}} \end{pmatrix}, \tag{57}$$

where we can think of 5D Lorentz indices breaking up into 4D Lorentz indices $A \mapsto (\alpha, \dot{\alpha})$ once a particular direction is chosen, means we manifestly break the 5D Lorentz symmetry.

The antisymmetry of p_{AB} implies the important identity that

$$\Omega^{AB}\lambda_A^{\ a}\lambda_B^{\ b} = 0, \tag{58}$$

so no Lorentz-invariant little-group tensors can be formed from just considering a single particle alone.

4.2 5D 3-particle Amplitude

Special Kinematics in 5D

Scattering amplitudes, are Lorentz invariant and transform covariantly under the little group. So like the massive 4D case, the amplitudes here are $\mathrm{SU}(2)$ covariants, means it carries $\mathrm{SU}(2)$ little group indices and no free Lorentz indices. We want keep the consistency with 4D so here the spin-s particle still transformed in the (2s+1)-dimension symmetric representation of 2s product of $\mathrm{SU}(2)$. In short, a spin-s particle should carry $2s$ $\mathrm{SU}(2)$ little group indices.

Then, a 3-particle amplitude of spin s_1, s_2, s_3 is a tensor carrying $(2s_1 + 2s_2 + 2s_3)$ indices like

$$\mathcal{M}_{(a_1 \cdots a_{2s_1})(b_1 \cdots b_{2s_2})(c_1 \cdots c_{2s_3})} \tag{59}$$

for a_i, b_j, c_k indices of $\mathrm{SU}(2)$ little group of particle 1, 2, 3 respectively.

In order to construct amplitudes which is a tensor carrying little group indices, it is necessary to figure out what tensors structure are possible here. The spinors that make up an on-shell 3-point amplitude without any Lorentz indices can be possibly listed as follows

$$y^{12}{}_{ab} := (\lambda_1)^A{}_a(\lambda_2)_{Ab} \tag{60}$$

$$y^{23}{}_{bc} := (\lambda_2)^A{}_b(\lambda_3)_{Ac} \tag{61}$$

$$y^{31}{}_{ca} := (\lambda_3)^A{}_c(\lambda_3)_{Aa} \tag{62}$$

Because of momentum conservation, the determinant of this y^{ij} vanishes

$$\mathrm{Det}(y^{12}) = \frac{1}{2}\epsilon^{ab}\epsilon^{cd}{y^{12}}_{ac}{y^{12}}_{bd} = \frac{1}{2}{y^{12}}_{ac}y^{12ac} = \frac{1}{2}p_1^{AB}p_{2AB} = 2(p_1 \cdot p_2) = 0 \tag{63}$$

This means we can decompose each of them into a outer product of $\mathrm{SU}(2)$ spinors:

$$\begin{aligned} {y^{12}}_{ab} &= {y^1}_a {y^2}_b &(64)\\ {y^{23}}_{bc} &= {y^2}_b {y^3}_c &(65)\\ {y^{31}}_{ca} &= {y^3}_c {y^1}_a &(66)\end{aligned}$$

In the following, we use the notation: $1_a := {y^1}_a$ and ${1^A}_a := {(\lambda_1)^A}_a$ for massless 5D spniors.

First, we can try to write down a 3-point amplitude using these variables, e.g. the Yukawa interaction in 5D is precisely $y^1 2_{ab}$:

$$\mathcal{M}_{a,b} = \text{Corresponding Feynman diagram} = g 1_a 2_b \tag{67}$$

Connecting to 4D x-factor

If there exist a similar ”x” factor in 5D, it would be of the form ${\chi^c}_{c'}$ with no Lorentz indices but little group indices. It can be understood as

$$p_2^{AB}{3_B}^c = {\chi^c}_{c'}3^{Ac'} \tag{68}$$

There is no inverse of χ because of $\chi^2 = 0$, $\mathrm{tr}\chi = 0$ and $\mathrm{Det}\chi = 0$. In the following, it can be known that

$$ {\chi^c}_{c'} = 3^c 3_{c'} \tag{69}$$

so the special kinematics $\mathrm{SU}(2)$ spinors ${y^3}_c = 3_c$ is a sort of “square root” of χ, the x-factor in 5D. Similarly, for other two particles, we can obtain the corresponding x-factor

$$\begin{aligned} p_3^{AB}{1_B}^a &= {\chi^a}_{a'}1^{Aa'} = (1_a 1^{a'})1^{Aa'} &(70)\\ p_1^{AB}{2_B}^b &= {\chi^b}_{b'}2^{Ab'} = (2_b 2^{b'})2^{Ab'} &(71)\end{aligned}$$

Then, in scalr-QED in 5D, the scalar-scalar-photon amplitude can be directly written as

$$\mathcal{M}_{cc'} = (\text{Corresponding Feynman diagram}) = g 3_c 3_{c'} \tag{72}$$

If one compactifies to 4D, all 4D massive amplitudes should satisfy $m_1 \pm m_2 \pm' m_3 = 0$. If one particle 3 is massless in 4D and other 2 are massive with same mass m, the 4D x factor appears in the form as

$$3_c 3_{c'} = \begin{pmatrix} -mx^{-1} & \mp im \\ \mp im & mx \end{pmatrix}_{cc'} \tag{73}$$

So we obtain the corresponding amplitude after compactifying, where the diagonal components refer to the massless 4-vector of particle 3, and the off-diagonal components represent the scalar sector. Either $\mathcal{M} \sim x$ or x^{-1} depending on its helicity.

5 Detailed Construction

We start from the convention for 5D gamma matrix and momentum

$$p^\mu = (p_0, \vec{p}, p^4), \quad \mu = \bar{\mu} = (0,1,2,3) \quad \mu = \hat{\mu} = 4 \tag{74}$$

$$p_\mu = (p^0, -\vec{p}, -p^4) \tag{75}$$

and

$$\Gamma^{\bar{\mu}} = \begin{pmatrix} 0 & \sigma^{\bar{\mu}} \\ \bar{\sigma}^{\bar{\mu}} & 0 \end{pmatrix} \tag{76}$$

$$\Gamma^4 = \begin{pmatrix} -i & & & \\ & -i & & \\ & & i & \\ & & & i \end{pmatrix} \tag{77}$$

where $\sigma^{\bar{\mu}} = (\mathbf{1}, \vec{\sigma})$ and $\bar{\sigma}^{\bar{\mu}} = (\mathbf{1}, -\vec{\sigma})$.

Then the Dirac operator can be written as

$$\begin{aligned} \not{p} &= p_\mu \Gamma^\mu = p_{\bar{\mu}} \Gamma^{\bar{\mu}} - p^4 \Gamma^4 \\ &= \begin{pmatrix} 0 & 0 & p^0 - p^3 & -p^1 + ip^2 \\ 0 & 0 & -p^1 - i - p^2 & p^0 + p^3 \\ p^0 + p^3 & p^1 - ip^2 & 0 & 0 \\ p^1 + ip^2 & p^0 - p^3 & 0 & 0 \end{pmatrix} - \begin{pmatrix} -ip^4 & & & \\ & -ip^4 & & \\ & & ip^4 & \\ & & & ip^4 \end{pmatrix} \\ &= \begin{pmatrix} ip^4 & 0 & p^0 - p^3 & -p^1 + ip^2 \\ 0 & ip^4 & -p^1 - i - p^2 & p^0 + p^3 \\ p^0 + p^3 & p^1 - ip^2 & -ip^4 & 0 \\ p^1 + ip^2 & p^0 - p^3 & 0 & -ip^4 \end{pmatrix} \end{aligned} \tag{78}$$

The determinant of this matrix equals to zero, which means it can be decomposed to the outer product of two 4-component spinor with $\mathrm{Sp}(2, \mathbb{C})$ little group indices

$$\not{p} = |p^a\rangle_A \langle p_a|^B \tag{79}$$

ant they satisfy the massless Dirac equation

$$\not{p} |p^a\rangle_A = 0. \tag{80}$$

A specific construction for the spinor-helicity variable is

$$|p^a\rangle_A = \begin{pmatrix} p^0 - p^3 & 0 \\ -p^1 - ip^2 & \dfrac{ip^4}{p^0 - p^3} \\ -ip^4 & \dfrac{p^1 - ip^2}{p^0 - p^3} \\ 0 & 1 \end{pmatrix} \tag{81}$$

and the bra can be obtained by introducing a symplectic form

$$\Omega_{AB} = \begin{pmatrix} 0 & -1 & & \\ 1 & 0 & & \\ & & 0 & -1 \\ & & 1 & 0 \end{pmatrix} = \Omega^{BA}. \tag{82}$$

Then the corresponding bra variable can be written like

$$\begin{aligned} \langle p_a|^A &= \epsilon_{ab}(\Omega^{AB}|p^b\rangle_B)^T \\ &= \begin{pmatrix} \frac{-ip^4}{p^0-p^3} & 0 & -1 & \frac{p^1-ip^2}{p^0-p^3} \\ -p^1-ip^2 & -p^0+p^3 & 0 & p^4 \end{pmatrix} \end{aligned} \tag{83}$$

Also, we have

$$\begin{aligned} |p_a\rangle_A &= \epsilon_{ab}|p^b\rangle_A \\ &= \begin{pmatrix} p^0-p^3 & 0 \\ -p^1-ip^2 & \dfrac{ip^4}{p^0-p^3} \\ -ip^4 & \dfrac{p^1-ip^2}{p^0-p^3} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & p^0-p^3 \\ \frac{-ip^4}{p^0-p^3} & -p^1-ip^2 \\ \frac{-p^1+ip^2}{p^0-p^3} & -ip^4 \\ -1 & 0 \end{pmatrix} \end{aligned} \tag{84}$$

Similar for 4d massless case, we can also construct the corresponding spinor-helicity variable from

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} p^0-p^3 & -p^1+ip^2 \\ -p^1-ip^2 & p^0+p^3 \end{pmatrix} = \lambda_\alpha\tilde{\lambda}_{\dot{\alpha}} \tag{85}$$

where the λ and $\tilde{\lambda}$ are represented by

$$\lambda_\alpha = |p\rangle_\alpha = \begin{pmatrix} p^0-p^3 \\ -p^1-ip^2 \end{pmatrix}, \qquad \tilde{\lambda}_{\dot{\alpha}} = [p|_{\dot{\alpha}} = \begin{pmatrix} 1 \\ \frac{-p^1+ip^2}{p^0-p^3} \end{pmatrix} \tag{86}$$

Also, we can obtain the corresponding "bra" and "ket" variable by contracting with antisymmetric tensor ϵ

$$\langle p|^\alpha = \epsilon^{\alpha\beta}|p\rangle_\beta = \begin{pmatrix} -p^1-ip^2 \\ -p^0+p^3 \end{pmatrix}, \qquad |p]^{\dot{\alpha}} = \epsilon^{\dot{\alpha}\dot{\beta}}[p|_{\dot{\beta}} = \begin{pmatrix} \frac{-p^1+ip^2}{p^0-p^3} \\ -1 \end{pmatrix} \tag{87}$$

Then it can found that if we choose the limit in which the fifth component $p^4 = 0$, the 5D spinor-helicity variable reduces to

$$\langle p_a|^A|_{p^4=0} = \begin{pmatrix} 0 & \langle p|^\alpha \\ -[p|_{\dot{\alpha}} & 0 \end{pmatrix}^T \tag{88}$$

and

$$|p_a\rangle_A|_{p^4=0} = \begin{pmatrix} 0 & |p\rangle_\alpha \\ |p]^{\dot\alpha} & 0 \end{pmatrix} \tag{89}$$

It is straightforward now to compute ${y^{ij}}_{ab}$, here we choose ${y^{12}}_{ab}$ as an example

$$\langle 1_a|2_b\rangle = \begin{pmatrix} [21] & 0 \\ 0 & \langle 12\rangle \end{pmatrix} = 1_a 2_b \tag{90}$$

From the 3 particle momentum conservation, we have known that 4D massless spinor-helicity variable should satisfy

$$|1\rangle \propto |2\rangle \propto |3\rangle \qquad \text{or} \qquad |1] \propto |2] \propto |3] \tag{91}$$

which means one of $[21]$ and $\langle 12\rangle$ will become vanishing.

First we consider non-vanishing $[21]$ case, in which we can choose a particular decomposition

$$y^{12} = 1_a 2_b = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} [21] \\ 0 \end{pmatrix} \tag{92}$$

$$y^{23} = \Upsilon_{2b} 3_c = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} [32] \\ 0 \end{pmatrix} \tag{93}$$

$$y^{31} = \Upsilon_{3c} \Upsilon_{1a} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} [13] \\ 0 \end{pmatrix} \tag{94}$$

from which we can read off

$$\Upsilon_{1a} = [13] 1_a, \quad \Upsilon_{2b} = \frac{1}{[21]} 2_b, \quad \Upsilon_{3c} = \frac{1}{[32]} 3_c \tag{95}$$

We denote these coefficients as A, B and C. We want to use the consistent decomposition for these y variables, means the same 2_b appearing both in y^{12} and y^{23}, like

$$y^{12} = 1'_a 2'_b \tag{96}$$
$$y^{23} = 2'_b 3'_c \tag{97}$$
$$y^{31} = 3'_c 1'_a \tag{98}$$

Because of the ambiguity in the decomposition, we can rescale like

$$1'_a = \alpha_1 1_a, \quad 2'_b = \alpha_2 2_b, \quad 3'_c = \alpha_3 3_c \tag{99}$$

which means the following equation should satisfy

$$y^{12} = \alpha_1 1_a \alpha_2 2_b = 1_a 2_b \tag{100}$$
$$y^{23} = \alpha_2 2_b \alpha_3 3_c = B 2_b C 3_c \tag{101}$$
$$y^{31} = \alpha_3 3_c \alpha_1 1_a = C 3_c A 1_a \tag{102}$$

So we have a group of equations

$$\alpha_1 \alpha_2 = 1 \tag{103}$$
$$\alpha_2 \alpha_3 = BC \tag{104}$$
$$\alpha_3 \alpha_1 = CA \tag{105}$$

and the solutions are given as following

$$\alpha_1 = \sqrt{\frac{AC}{B}}, \quad \alpha_2 = \sqrt{\frac{B}{AC}}, \quad \alpha_3 = \sqrt{ABC} \tag{106}$$

After inserting these solutions, we obtain the final decomposition

$$1_a = \sqrt{\frac{[13][21]}{[32]}}\begin{pmatrix}1\\0\end{pmatrix}, \quad 2_b = \sqrt{\frac{[21][32]}{[13]}}\begin{pmatrix}1\\0\end{pmatrix}, \quad 3_c = \sqrt{\frac{[13][32]}{[21]}}\begin{pmatrix}1\\0\end{pmatrix}. \tag{107}$$

In order to give the correct Yang-Mills amplitude, we need the ”inverse” of these two-component spinor, defined as following

$$\hat{i}^a i_a = 1. \tag{108}$$

From (107), we can explicitly solve the inverse spinor

$$\hat{1}^a = \sqrt{\frac{[32]}{[13][21]}}\begin{pmatrix}1\\0\end{pmatrix} + \alpha_1 1^a, \tag{109}$$

$$\hat{2}^b = \sqrt{\frac{[13]}{[21][32]}}\begin{pmatrix}1\\0\end{pmatrix} + \alpha_2 2^b, \tag{110}$$

$$\hat{3}^c = \sqrt{\frac{[21]}{[13][32]}}\begin{pmatrix}1\\0\end{pmatrix} + \alpha_3 3^c \tag{111}$$

here the second means the shift ambiguity. (α are arbitrary complex number here)

Because of the physical constraint

$$\Lambda \cdot z = 0 \tag{112}$$

we need

$$\hat{\delta}_{1A} + \hat{\delta}_{2A} + \hat{\delta}_{3A} + (\alpha_1 + \alpha_2 + \alpha_3)\delta_A = 0 \tag{113}$$

where $\hat{\delta}_{iA} = \hat{i}^a i_{Aa}$ and $\delta_A = i^a i_{Aa}$.

We also know that

$$\hat{\delta}_{1A} + \hat{\delta}_{2A} + \hat{\delta}_{3A} = \hat{K}\delta_A \tag{114}$$

so in order to satisfy the constraint, we just need

$$\hat{K} + (\alpha_1 + \alpha_2 + \alpha_3) = 0 \tag{115}$$

Here, we can try to give the explicit form of $\hat{\delta}_{1A}$

$$\hat{\delta}_{1A} = \hat{1}^a 1_{Aa} = \begin{pmatrix} 0 & p^0 - p^3 \\ 0 & -p^1 - ip^2 \\ \frac{-p^1+ip^2}{p^0-p^3} & 0 \\ -1 & 0 \end{pmatrix} \sqrt{\frac{[32]}{[13][21]}}\begin{pmatrix}1\\0\end{pmatrix} = \sqrt{\frac{[32]}{[13][21]}}\begin{pmatrix} 0 \\ 0 \\ \frac{-p^1+ip^2}{p^0-p^3} \\ -1 \end{pmatrix} = \sqrt{\frac{[32]}{[13][21]}}\begin{pmatrix} 0 \\ |1]^{\dot{\alpha}} \end{pmatrix} \tag{116}$$

The non-vanishing component can be written as

$$\sqrt{\frac{[32]}{[13][21]}}|1]^{\dot{\alpha}} = \sqrt{\frac{1}{[32][21][13]}}[32]|1]^{\dot{\alpha}} \tag{117}$$

From the Schouten identity

$$|i]\langle jk\rangle + |k]\langle ij\rangle + |j]\langle ki\rangle = 0 \tag{118}$$

we can obtain

$$[32]|1] = -([13]|2] + [21]|3]). \tag{119}$$

Surprisingly, we can find that

$$\sqrt{\frac{1}{[32][21][13]}}[13]|2] \qquad \text{and} \qquad \sqrt{\frac{1}{[32][21][13]}}[21]|3] \tag{120}$$

are just the non-vanishing components of $\hat{\delta}_{2A}$ and $\hat{\delta}_{3A}$, in other words,

$$\hat{\delta}_{1A} = -\hat{\delta}_{2A} - \hat{\delta}_{3A} \tag{121}$$

so that

$$\hat{\delta}_{1A} + \hat{\delta}_{2A} + \hat{\delta}_{3A} = 0 \tag{122}$$

which means the constraint is equivalent to

$$(\alpha_1 + \alpha_2 + \alpha_3)\delta_A = 0 \quad \Leftrightarrow \quad \alpha_1 + \alpha_2 + \alpha_3 = 0 \tag{123}$$

Here, we can simply choose them to be 0. Also, we can confirm $\delta_A = 1^a 1_{Aa} = 2^b 2_{Ab} = 3^c 3_{Ac}$ through these explicit forms

$$1^a 1_{Aa} = \begin{pmatrix} 0 & p^0 - p^3 \\ 0 & -p^1 - ip^2 \\ \frac{-p^1 + ip^2}{p^0 - p^3} & 0 \\ -1 & 0 \end{pmatrix} \sqrt{\frac{[13][21]}{[32]}} \begin{pmatrix} 0 \\ -1 \end{pmatrix} = \sqrt{\frac{[13][21]}{[32]}} \begin{pmatrix} -(p^0 - p^3) \\ -(-p^1 - ip^2) \\ 0 \\ 0 \end{pmatrix} = \sqrt{\frac{[13][21]}{[32]}} \begin{pmatrix} -|1\rangle_\alpha \\ 0 \end{pmatrix} \tag{124}$$

From the momentum conservation

$$|1\rangle[1| + |2\rangle[2| + |3\rangle[3| = 0 \tag{125}$$

by contracting a $|3]$ in both sides, we can obtain

$$|1\rangle = -\frac{[23]}{[13]}|2\rangle \tag{126}$$

and this gives us the following result

$$\sqrt{\frac{[13][21]}{[32]}} \begin{pmatrix} \frac{[23]}{[13]}|2\rangle_\alpha \\ 0 \end{pmatrix} = \sqrt{\frac{[21][32]}{[13]}} \begin{pmatrix} -|2\rangle_\alpha \\ 0 \end{pmatrix} = 2^b 2_{Ab} \tag{127}$$

So we confirm the fact

$$\delta_A = 1^a 1_{Aa} = 2^b 2_{Ab} \tag{128}$$

the remaining two can be confirmed in similar manner.

Suppose we want to compute the amplitude with two spin $\frac{1}{2}$ fermions and spin 1 gauge boson in 5D, the amplitude can be directly written as

$$\mathcal{M}_{ab(cc')} = g(1_a 2_b \hat{3}_c + 1_a \hat{2}_b 3_c + \hat{1}_a 2_b 3_c) 3_{c'}. \tag{129}$$

By using the explicit form of these building blocks, it can written as

$$\begin{aligned}\mathcal{M}_{ab(cc')} = &\left(\sqrt{\frac{[13][21]}{[32]}}\begin{pmatrix}1\\0\end{pmatrix}_a \sqrt{\frac{[21][32]}{[13]}}\begin{pmatrix}1\\0\end{pmatrix}_b \sqrt{\frac{[21]}{[13][32]}}\begin{pmatrix}0\\1\end{pmatrix}_c + \right.\\ &\sqrt{\frac{[13][21]}{[32]}}\begin{pmatrix}1\\0\end{pmatrix}_a \sqrt{\frac{[13]}{[21][32]}}\begin{pmatrix}0\\1\end{pmatrix}_b \sqrt{\frac{[13][32]}{[21]}}\begin{pmatrix}1\\0\end{pmatrix}_c + \\ &\left.\sqrt{\frac{[32]}{[13][21]}}\begin{pmatrix}0\\1\end{pmatrix}_a \sqrt{\frac{[21][32]}{[13]}}\begin{pmatrix}1\\0\end{pmatrix}_b \sqrt{\frac{[13][32]}{[21]}}\begin{pmatrix}1\\0\end{pmatrix}_c\right) \sqrt{\frac{[13][32]}{[21]}}\begin{pmatrix}1\\0\end{pmatrix}_{c'}\end{aligned} \tag{130}$$

By picking the different indices, we can reproduce the different helicity amplitudes in 4D, like

$$a, b, (c, c') = 1, 2, (1, 1) \quad \Rightarrow \quad \sqrt{\frac{[13][21]}{[32]}}\sqrt{\frac{[13]}{[21][32]}}\frac{[13][32]}{[21]} = \frac{[13]^2}{[21]} \tag{131}$$

$$a, b, (c, c') = 1, 1, (1, 2) = \sqrt{\frac{[13][21]}{[32]}}\sqrt{\frac{[32][21]}{[13]}}\frac{1}{2}(1+1) = [21] \tag{132}$$

$$a, b, (c, c') = 2, 1, (1, 1) = \sqrt{\frac{[32]}{[13][21]}}\sqrt{\frac{[21][32]}{[13]}}\frac{[13][32]}{[21]} = \frac{[32]^2}{[21]} \tag{133}$$

From the known 4D results

$$M^{h_1 h_2 h_3} = \begin{cases} \tilde{g}\,[12]^{h_1+h_2-h_3}\,[23]^{h_2+h_3-h_1}\,[31]^{h_3+h_1-h_2}, & \text{when } h_1+h_2+h_3 > 0, \\ g\,\langle 12\rangle^{h_3-h_1-h_2}\,\langle 23\rangle^{h_1-h_2-h_3}\,\langle 31\rangle^{h_2-h_3-h_1}, & \text{when } h_1+h_2+h_3 < 0. \end{cases} \tag{134}$$

we know that the 3 point amplitudes with $(1/2^+, 1/2^-, +1)$ configuration equals to

$$\mathcal{M}(+\frac{1}{2}, -\frac{1}{2}, +1) = \frac{[31]^2}{[12]} \tag{135}$$

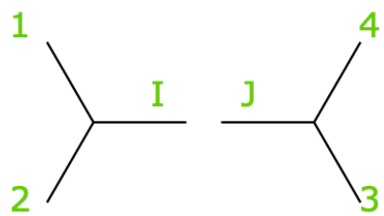
$$\mathcal{M}(-\frac{1}{2}, +\frac{1}{2}, +1) = \frac{[32]^2}{[12]} \tag{136}$$

$$\mathcal{M}(0, 0, +1) = [12] \tag{137}$$

We found that the 5D 3 point amplitudes actually reproduce the corresponding 4D amplitudes.

6 5D 4-point amplitude

Consider the s-channel limit of a 4-point scattering amplitude.



$$\rightarrow \quad \frac{1}{s} M_3(1^a_A, 2^b_A, I^e_A) M_3(J^e_A, 3^c_A, 4^d_A) \tag{138}$$

In this case, the propagator $P_I^{AB} = -P_J^{AB}$ is on-shell because of

$$P_I^{AB} P_{I\,AB} = 2(p_1 + p_2)^2 = 2s = 0 \tag{139}$$

which means that we can use the 3 particle special kinematics what we learned before.

For the internal line, we have

$$P_{I\,AB} = {I_A}^e I_{B\,e} = -{J_A}^e J_{B\,e} = -P_{J\,AB} \tag{140}$$

$$\Rightarrow {J_A}^e = \pm i {I_A}^e \tag{141}$$

here we choose the convention ${J_A}^e = +i{I_A}^e$.

For the two 3 point subamplitude, we can use the special kinematics as following

$$\begin{cases} {y^{12}}_{ab} = {1^A}_a 2_{Ab} = 1_a 2_b & {y^{J3}}_{ec} = {J^A}_e 3_{Ac} = +I_e 3_c = J_e 3_c \\ {y^{2I}}_{be} = {2^A}_b I_{Ae} = 2_b I_e & {y^{34}}_{cd} = {3^A}_c 4_{Ad} = 3_c 4_d \\ {y^{I1}}_{ea} = {I^A}_e 1_{Aa} = I_e 1_a & {y^{4J}}_{de} = {4^A}_d i I_{Ae} = 4_d J_e \end{cases} \tag{142}$$

where the hat variable is defined like before

$$\hat{i}^a i_a = 1 \quad \Leftrightarrow \quad \hat{i}_{[a} i_{b]} = \frac{1}{2}\epsilon_{ab}. \tag{143}$$

Also from the 3-point momentum conservation, we have

$$\delta_A := 1^a 1_{Aa} = 2^b 2_{Ab} = I^e I_{Ae} \tag{144}$$

$$\delta'_A := 3^c 3_{Ac} = 4^d 4_{Ad} = J^e i I_{Ae} \tag{145}$$

$\star$ Notice that while $J_{Ae} = +iI_{Ae}$, $J_e \neq I_e$. ($I^e J_e$ is important and non-trival in the 4-point amplitude.) And don't forget that here we have the shift redundancy $\alpha_1 + \alpha_2 + \alpha_I = 0$ and $\alpha_3 + \alpha_4 + \alpha_J = 0$.

In fact. on the s-pole, we have a important relation

$$I^e J_e = \sqrt{-u} = \sqrt{t} \qquad (\text{when } s = 0, t = -u) \tag{146}$$

Proof: Consider $y^{13}{}_{ac} = 1^A{}_a 3_{Ac}$

$$\det y^{13}{}_{ac} = \frac{1}{2}\epsilon^{aa'}\epsilon^{cc'}y^{13}{}_{ac}y^{13}{}_{a'c'} = \frac{1}{2}1^A{}_a 3_{Ac}\epsilon^{aa'}\epsilon^{cc'}1^B{}_{a'}3_{Bc'}$$
$$= \frac{1}{2}1^A{}_a 3_{Ac}1^{Ba}3_B{}^c = \frac{1}{2}\langle 1_a|3_c\rangle\langle 1^a|3^c\rangle = 2p_1\cdot p_3 = u \tag{147}$$

where $u = (p_1+p_3)^2 = p_1^2 + p_3^2 + 2p_1\cdot p_3 = 2p_1\cdot p_3$.

A key obervation here is that $y^{13}{}_{ac}y^{13}{}_a{}^{c'}$ is antisymmetric in $c \leftrightarrow c'$, and $y^{13}{}_{ac}y^{13}{}_{a'}{}^c$ is antisymmetric in $a \leftrightarrow a'$ (proof is skipped).

From the equation (143), we can rewrite u in three different ways

$$\begin{aligned}
1.\quad u &= -\hat{1}^{[a}1^{a']}y^{13}{}_{ac}y^{13}{}_{a'}{}^c \\
&= -\hat{1}^{a}1^{a'}y^{13}{}_{ac}y^{13}{}_{a'}{}^c \\
&= -\hat{\delta}_1^A\delta^B 3_{Ac}3_B{}^c \\
&= -\hat{\delta}_1^A(I^eI^B{}_e)3_{Ac}3_B{}^c \\
&= -\hat{\delta}_1^A I^e(-iJ_e3^c)3_{Ac} \\
&= +i(I^eJ_e)\hat{\delta}_1^A\delta'_A
\end{aligned} \tag{148}$$

$$\begin{aligned}
2.\quad u &= -\hat{3}^{[c}3^{c']}y^{13}{}_{ac}y^{13\,a}{}_{c'} \\
&= -\hat{3}^{c}3^{c'}y^{13}{}_{ac}y^{13\,a}{}_{c'} \\
&= -\hat{3}^{c}3^{c'}1^A{}_a3_{Ac}1^{Ba}3_{Bc'} \\
&= -\hat{\delta}_{3A}(J^eiI_{Be})1^{Ba}1^A{}_a \\
&= -\hat{\delta}_{3A}J^e(-i1^aI_e)1^A{}_a \\
&= -i(I^eJ_e)\delta^A\hat{\delta}_{3A}
\end{aligned} \tag{149}$$

$$\begin{aligned}
3.\quad u &= 2\times\hat{1}^{[a}1^{a']}\hat{3}^{[c}3^{c']}y^{13}{}_{ac}y^{13}{}_{a'c'} \\
&= 2\times\hat{1}^{[a}1^{a']}\hat{3}^{[c}3^{c']}1^A{}_a3_{Ac}1^B{}_{a'}3_{Bc'} \\
&= 2\times\hat{\delta}_1^{[A}\delta^{B]}\hat{\delta}_{3\,[A}\delta'{}_{B]} \\
&= -\hat{\delta}_1{}^B\delta^A\hat{\delta}_{3A}\delta'_B \\
&= -(\delta^A\hat{\delta}_{3A})(\hat{\delta}_1{}^B\delta'_B)
\end{aligned} \tag{150}$$

$$1\times 2 \quad\Rightarrow\quad u^2 = (I^eJ_e)^2(\delta^A\hat{\delta}_{3A})(\hat{\delta}_1{}^B\delta'_B) = (I^eJ_e)^2(-u) \tag{151}$$
$$\Longrightarrow I^eJ_e = \sqrt{-u} \tag{152}$$

So we finish the proof. This also implies that

$$\hat{\delta}_1^A\delta'_A = -\sqrt{u} \qquad \text{and} \qquad \delta^A\hat{\delta}_{3A} = \sqrt{u}. \tag{153}$$

From $\hat{I}^eI^f\hat{J}_{[e}J_{f]} = \frac{1}{2}\hat{I}^eI^f\epsilon_{ef} = \frac{1}{2}$, what we can do next is expanding in both sides

$$(\hat{I}^e\hat{J}_e)(I^fJ_f) - (\hat{I}^eJ_e)(I^f\hat{J}_f) = 1 \tag{154}$$

because of the redundancy of definition of the hat variables, we can shift the definition of $\hat{I}$ and $\hat{J}$ to impose the condition $\hat{I}^e J_e = 0 = I^f \hat{J}_f$, so that we have

$$(\hat{I}^e \hat{J}_e)(I^f J_f) = 1$$
$$\Rightarrow \quad \hat{I}^e \hat{J}_e = \frac{1}{I^f J_f} = \frac{1}{\sqrt{-u}} \tag{155}$$

Before, for the on-shell 3-particle kinematics, we can decompose the corresponding y variable into the outer product of two no-hatted variables. But we didn't address with the y variable like y^{13}, because its leg crosses through two on-shell 3-point subamplitudes, so that we cann't utilize the special kinematics. Fortunately, for every leg, not ony i variable, we also have $\hat{i}$ variable, which combined together to expand a basis for every little group space. It means that in general, we can expand y^{13} as following

$$y^{13}{}_{ac} = X 1_a 3_c + Y \hat{1}_a 3_c + Z 1_a \hat{3}_c + W \hat{1}_a \hat{3}_c \tag{156}$$

then contracting with $1^a \hat{3}^c$, we can obtain one coefficient

$$1^a \hat{3}^c y^{13}{}_{ac} = -Y = 1^a \hat{3}^c 1^A{}_a 3_{Ac} = \delta^A \hat{\delta}_{3A} = \sqrt{u} \tag{157}$$

By contracting $\hat{1}^a 3^c$ and $1^a 3^c$, we can obtain the corresponding coefficient

$$-Z = \hat{\delta}_1^A \delta'_A = -\sqrt{u} \tag{158}$$
$$W = \delta^A \delta'_A = 0 \tag{159}$$

Thus

$$y^{13}{}_{ac} = X 1_a 3_c - \sqrt{u} \hat{1}_a 3_c + \sqrt{u} 1_a \hat{3}_c. \tag{160}$$

Till now, we cannot fix the constant X, but it will be completly fixed later.

We can get the expansion of y^{ij} quickly and systematically by finding the various contractions of δ_A and $\hat{\delta}_{iA}$. First notice that $\delta_A \delta'^A = 0$ and any objects contracted with itself gives the vanishing result(e.g. $\delta^A \delta_A = 1^a 1^A{}_a 1^b 1_{Ab} = 1^a 1^b \langle 1_a | 1_b \rangle = 0$).

One can also check the following contractions give vanishing results

$$\delta^A \hat{\delta}_{iA} = 0 \qquad \delta'^A \hat{\delta}_{jA} = 0 \tag{161}$$
$$\hat{\delta}_i{}^A \hat{\delta}_{i'A} = 0 \qquad \hat{\delta}_j{}^A \hat{\delta}_{j'A} = 0 \tag{162}$$

for $i, i' \in \{1, 2, I\}$ and $j, j' \in \{3, 4, J\}$.

For example,

$$\delta^A \hat{\delta}_{iA} = i^a i^A{}_a \hat{i}^b i_{Ab} = i^a \hat{i}^b \langle i_a | i_b \rangle = 0 \tag{163}$$
$$\hat{\delta}_i{}^A \hat{\delta}_{i'A} = \hat{i}^a i^A{}_a \hat{i'}^b i'_{Ab} = \hat{i}^a \hat{i'}^b y^{ii'}{}_{ab} = \hat{i}^a \hat{i'}^b i_a i'_b = 1\,?? \tag{164}$$
$$\hat{\delta}_1^A \hat{\delta}_{IA} = \hat{1}^a 1_a^A \hat{I}^b I_{Ab} \tag{165}$$

Any contraction involing particles I, J gives 0, so the remaining possible contractions are

$$\delta'^A \hat{\delta}_{iA} \qquad \delta^A \hat{\delta}'_{jA} \qquad \hat{\delta}_i^A \hat{\delta}_{jA} \tag{166}$$

for $i \in \{1,2\},\ j \in \{3,4\}$.

We use the 4 particle momentum conservation to help us obtain some useful identities

$$0 = \epsilon_{ab} 1_A{}^a 1_B{}^b + \epsilon_{ab} 2_A{}^a 2_B{}^b + \epsilon_{ab} 3_A{}^a 3_B{}^b + \epsilon_{ab} 4_A{}^a 4_B{}^b \tag{167}$$

$$\implies 0 = \hat{1}_{[a} 1_{b]} 1_A{}^a 1_B{}^b + \hat{2}_{[a} 2_{b]} 2_A{}^a 2_B{}^b + \hat{3}_{[a} 3_{b]} 3_A{}^a 3_B{}^b + \hat{4}_{[a} 4_{b]} 4_A{}^a 4_B{}^b \tag{168}$$

$$= \hat{1}_a 1_b 1_{[A}{}^a 1_{B]}{}^b + \hat{2}_a 2_b 2_{[A}{}^a 2_{B]}{}^b + \hat{3}_a 3_b 3_{[A}{}^a 3_{B]}{}^b + \hat{4}_a 4_b 4_{[A}{}^a 4_{B]}{}^b \tag{169}$$

$$0 = (\hat{\delta}_{1[A} + \hat{\delta}_{2[A}) \delta_{B]} + (\hat{\delta}_{3[A} + \hat{\delta}_{4[A}) \delta'_{B]} \tag{170}$$

Contracting with δ^A gives

$$\begin{aligned} 0 &= \delta^A (\hat{\delta}_{1[A} + \hat{\delta}_{2[A}) \delta_{B]} + \delta^A (\hat{\delta}_{3[A} + \hat{\delta}_{4[A}) \delta'_{B]} \\ &= \delta^A (\hat{\delta}_{3[A} + \hat{\delta}_{4[A}) \delta'_{B]} \\ &= \delta^A (\hat{\delta}_{3A} + \hat{\delta}_{4A}) \delta'_B \end{aligned} \tag{171}$$

which implies to

$$0 = \delta^A (\hat{\delta}_{3A} + \hat{\delta}_{4A}). \tag{172}$$

Contracting with δ'^A gives

$$0 = \delta'^A (\hat{\delta}_{1A} + \hat{\delta}_{2A}) \tag{173}$$

$$\vdots \tag{174}$$

Aftering fixing all special y_a spinors, we can conclude that

$$y^{12}{}_{ab} = 1_a 2_b \tag{175}$$

$$y^{13}{}_{ac} = \sqrt{u} \left(1_a \hat{3}_c - \hat{1}_a 3_c \right) \tag{176}$$

$$y^{14}{}_{ad} = \sqrt{u} \left(1_a \hat{4}_d + \hat{1}_a 4_d \right) \tag{177}$$

$$y^{23}{}_{bc} = \sqrt{u} \left(-2_b \hat{3}_c - \hat{2}_b 3_c \right) \tag{178}$$

$$y^{24}{}_{bd} = \sqrt{u} \left(-2_b \hat{4}_d + \hat{2}_b 4_d \right) \tag{179}$$

$$y^{34}{}_{cd} = 3_c 4_d \tag{180}$$

We can use these identities to form other quantities like

$$y^{12a}{}_b \, y^{13}{}_{ac} = \sqrt{u} \, 2_b 3_c. \tag{181}$$

which can help us to significantly simplify the computation for the 4-point amplitude.

Notice that the y_a spinors provide a more efficient description of kinematics subject to momentum conservation than the conventional spinor–helicity bilinears $y^{ij}{}_{ab}$, since the latter admit many redundant representations of structures such as $2_b 3_c$. For instance,

$$y^{42d}{}_b \, y^{43}{}_{dc} \;\sim\; 2_b 3_c \;\sim\; y^{42d}{}_b \, y^{43}{}_d{}^{c'} \, y^{13}{}_{ac'} \, y^{13a}{}_c. \tag{182}$$

Nevertheless, the residues of the full amplitude at its poles must ultimately be expressible solely in terms of the standard spinor–helicity pairs $y^{ij}{}_{ab}$.

6.1 Example of 4-point amplitudes

Yang – Mills Considering we glue two 3-point YMs amplitudes, so that we obtain the s pole residue

$$\lim_{s\to 0} s\,\mathcal{M}_4 = \mathcal{M}^{(ee)}_{3\,(aa)(bb)}(1,2,I)\,\mathcal{M}_{3\,(ee)(cc)(dd)}(J,3,4) \tag{183}$$

$$= g^2\Big[(\hat{1}_a 2_b I^e + 1_a\hat{2}_b I^e + 1_a 2_b \hat{I}^e)(\hat{J}_e 3_c 4_d + J_e\hat{3}_c 4_d + J_e 3_c \hat{4}_d)\Big]^2 \tag{184}$$

$$= g^2\left[\sqrt{-u}\,(\hat{1}_a 2_b + 1_a\hat{2}_b)(\hat{3}_c 4_d + 3_c\hat{4}_d) + \frac{1}{\sqrt{-u}}\,(1_a 2_b 3_c 4_d)\right]^2 \tag{185}$$

$$= \frac{g^2}{-u}\Big[-u\,(\hat{1}_a 2_b + 1_a\hat{2}_b)(\hat{3}_c 4_d + 3_c\hat{4}_d) + 1_a 2_b 3_c 4_d\Big]^2. \tag{186}$$

The square here means the symmetric tensor product of little group indices. Following the previous dictionary, we can see that $y^{12}y^{34} = 1234$, where we use the abbreviation $1234 = 1_a 2_b 3_c 4_d$. And also

$$\begin{aligned} y^{14}y^{23} &= u\,(1\hat{4} + \hat{1}4)(-2\hat{3} - \hat{2}3) \\ &= -u\,(12\hat{3}\hat{4} + 1\hat{2}3\hat{4} + \hat{1}2\hat{3}4 + \hat{1}\hat{2}34), \end{aligned} \tag{187}$$

$$\begin{aligned} y^{13}y^{24} &= u\,(1\hat{3} - \hat{1}3)(-2\hat{4} + \hat{2}4) \\ &= u\,(-12\hat{3}\hat{4} + 1\hat{2}\hat{3}4 + \hat{1}23\hat{4} - \hat{1}\hat{2}34). \end{aligned} \tag{188}$$

So the combination

$$y^{12}y^{34} - y^{13}y^{24} + y^{14}y^{23} = 1234 - u(\hat{1}2 + 1\hat{2})(\hat{3}4 + 3\hat{4}) \tag{189}$$

Therefore we obtain the s-pole residue

$$\lim_{s\to 0} s\,\mathcal{M}_4 = \frac{g^2}{t}\left[{y^{12}}_{ab}{y^{34}}_{cd} - {y^{13}}_{ac}{y^{24}}_{bd} + {y^{14}}_{ad}{y^{23}}_{bc}\right]^2. \tag{190}$$

Now the entire residue is written in terms of general spinors that don't just live on the pole $s = 0$. As we expected, a single pole in t channel appearing in the s-pole residue. This evokes the existence of another factorization channel in order for the YM amplitude to be consistent.

In order to obtain the t-pole residue, we only need to exchange $2 \leftrightarrow 4$ and $s \leftrightarrow t$. Notice that under $2 \leftrightarrow 4$ exchange,

$${y^{12}}_{ab}{y^{34}}_{cd} - {y^{13}}_{ac}{y^{24}}_{bd} + {y^{14}}_{ad}{y^{23}}_{bc} \to {y^{14}}_{ad}{y^{32}}_{cb} - {y^{13}}_{ac}{y^{42}}_{db} + {y^{12}}_{ab}{y^{43}}_{dc} \tag{191}$$

which means it keeps invariant under crossing. So we can conclude that

$$\lim_{t\to 0} t\,\mathcal{M}_4 = \frac{g^2}{s}\left[{y^{12}}_{ab}{y^{34}}_{cd} - {y^{13}}_{ac}{y^{24}}_{bd} + {y^{14}}_{ad}{y^{23}}_{bc}\right]^2. \tag{192}$$

Similar to u-pole, which need us to do $2 \leftrightarrow 3$ and $s \leftrightarrow u$, then

$$\lim_{u \to 0} u\, \mathcal{M}_4 = \frac{g^2}{t} \left[{y^{12}}_{ab} {y^{34}}_{cd} - {y^{13}}_{ac} {y^{24}}_{bd} + {y^{14}}_{ad} {y^{23}}_{bc} \right]^2 . \tag{193}$$

From the pole structure, we can make the ansatz for full 4-point amplitude, like

$$\mathcal{M}_4 = g^2 \left[{y^{12}}_{ab} {y^{34}}_{cd} - {y^{13}}_{ac} {y^{24}}_{bd} + {y^{14}}_{ad} {y^{23}}_{bc} \right]^2 \left(\frac{A_{a_1a_2a_3a_4}}{st} + \frac{B_{a_1a_2a_3a_4}}{tu} + \frac{C_{a_1a_2a_3a_4}}{su} \right) \tag{194}$$

where the constants $A_{a_1a_2a_3a_4}, B_{a_1a_2a_3a_4}, C_{a_1a_2a_3a_4}$ are related to color factor.

In order to make the full amplitude consistent, the ansatz should satisfy the following condition

$$\begin{aligned} \operatorname*{Res}_{s=0} \mathcal{M}_4 &= g^2 \frac{\left[{y^{12}}_{ab} {y^{34}}_{cd} - {y^{13}}_{ac} {y^{24}}_{bd} + {y^{14}}_{ad} {y^{23}}_{bc} \right]^2}{t} (A_{a_1a_2a_3a_4} - C_{a_1a_2a_3a_4}) \\ &= g^2 \frac{\left[{y^{12}}_{ab} {y^{34}}_{cd} - {y^{13}}_{ac} {y^{24}}_{bd} + {y^{14}}_{ad} {y^{23}}_{bc} \right]^2}{t} c_s \end{aligned} \tag{195}$$

and for other two channels, we have the similar condition, which combine together to give us

$$C - A = c_s, \quad A - B = c_t, \quad B - C = c_u. \tag{196}$$

We can solve for A, B, C if and only if the color factor c_s, c_t, c_u satisfy the following identities

$$c_s + c_t + c_u = 0 \tag{197}$$

which is just the Jacobbi identity.

We can choose a special solution to this group of equation

$$C = 0, \quad A = -c_s, \quad B = c_u. \tag{198}$$

Now the full amplitude are written like

$$\mathcal{M}_4 = g^2 \left[{y^{12}}_{ab} {y^{34}}_{cd} - {y^{13}}_{ac} {y^{24}}_{bd} + {y^{14}}_{ad} {y^{23}}_{bc} \right]^2 \left(\frac{-c_s}{st} + \frac{c_u}{tu} \right) \tag{199}$$

For simplicity, here we only compute the color-ordered amplitude. The important ingredient is the term inside the square parentheses, which includes many terms. In the following, we will analyse these terms carefully.

In the following, we only consider a special momentum choice, for which all of the fifth component are chosen to be 0. From the definition of y variable

$$y^{ij} = \lambda^A_{ia} \lambda_{ibA} = \langle i | j \rangle \tag{200}$$

it behaves as a diagonal 2×2 matrix, e.g. y^{12} is

$${y^{12}}_{ab} = \begin{pmatrix} [21] & \\ & \langle 12 \rangle \end{pmatrix} \tag{201}$$

So the term we focus on looks like

$$\left[\begin{pmatrix}[21] & \\ & \langle 12\rangle\end{pmatrix}\begin{pmatrix}[43] & \\ & \langle 34\rangle\end{pmatrix}-\begin{pmatrix}[31] & \\ & \langle 13\rangle\end{pmatrix}\begin{pmatrix}[42] & \\ & \langle 24\rangle\end{pmatrix}+\begin{pmatrix}[41] & \\ & \langle 14\rangle\end{pmatrix}\begin{pmatrix}[32] & \\ & \langle 23\rangle\end{pmatrix}\right]^2 \tag{202}$$

Because we are considering the pure YMs amplitude, every external particle are accompanied with two little group indices, which runs from 1 to 2. Here we use a notation (aa') to represent the indices configuration for a specific external particle. For example,

$$(11)(11)(11)(11) \tag{203}$$

means we choose $a, a', b, b', c, c', d, d' = 1$, and because of the special choice of momentum, most of the components give vanishing result. Here we only show the nonvanishing results.

The first one is $(11)(11)(11)(11)$, which gives

$$\begin{aligned}(11)(11)(11)(11) \quad \Rightarrow \quad & [21]^2[43]^2 + [31]^2[42]^2 + [41]^2[32]^2 - 2[21][42][31][42] \\ & - 2[31][42][41][32] + 2[21][43][41][32] \\ & = ([21][43] - [31][42] + [41][32])^2\end{aligned} \tag{204}$$

From the Fierz identity, we know that

$$[21][43] + [23][14] + [24][31] = 0. \tag{205}$$

So the equation (204) equals to 0, which is expected for 4D color ordered amplitude $A_4[1^+, 2^+, 3^+, 4^+] = 0$. For the similar reason, the component $(22)(22)(22)(22)$ also equals to 0, with all square brackets flipping to angle brackets.

The component $(22)(11)(11)(11)$ directly gives us 0 result, which is also expected for 4D color-ordered amplitude $A_4[1^-, 2^+, 3^+, 4^+] = 0$. And similar for $(11)(22)(11)(11)$, $(11)(11)(22)(11)$, $(11)(11)(11)(22)$.

The next nonvanishing component is $(22)(22)(11)(11)$, which gives

$$(22)(22)(11)(11) \quad \Rightarrow \quad \langle 12\rangle^2[43]^2. \tag{206}$$

Then the color-ordered amplitude can be computed as

$$\frac{\langle 12\rangle^2[43]^2}{st} = \frac{\langle 12\rangle^2[43]^2}{\langle 12\rangle[21]\langle 23\rangle[32]} \tag{207}$$

From the momentum conservation,we can obtain a relation for $[43]$

$$\sum |i\rangle[i| = 0 \quad \Rightarrow \quad \langle 1|(\sum |i\rangle[i|)|3] = 0 \tag{208}$$

which implies

$$\langle 12\rangle[23] + \langle 14\rangle[43] = 0 \quad \Rightarrow \quad [43] = -\frac{\langle 12\rangle}{\langle 14\rangle}[32] \tag{209}$$

So the amplitude equals to

$$\frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\langle 14\rangle^2}\times\frac{[32]}{[21]} \tag{210}$$

Again from momentum coservation, we can obtain

$$\frac{[32]}{[21]}=\frac{\langle 41\rangle}{\langle 34\rangle} \tag{211}$$

which combine together to give us

$$\frac{\langle 12\rangle^4}{\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 41\rangle} \tag{212}$$

This is just the Parke-Talyor foumula for color-ordered amplitude $A_4[1^-,2^-,3^+,4^+]$.

We can also check the second partial amplitude

$$\frac{\langle 12\rangle^2[43]^2}{tu}=\frac{\langle 12\rangle^2[43]^2}{\langle 13\rangle[31]\langle 32\rangle[23]} \tag{213}$$

After inserting

$$[43]=-\frac{\langle 12\rangle}{\langle 14\rangle}[32] \quad \text{and} \quad \frac{[23]}{[31]}=\frac{\langle 41\rangle}{\langle 42\rangle} \tag{214}$$

we can obtain the partial amplitude

$$\frac{\langle 12\rangle^2}{\langle 13\rangle\langle 32\rangle}\times\frac{\langle 12\rangle^2}{\langle 14\rangle^2}\times\frac{\langle 41\rangle}{\langle 42\rangle}=\frac{\langle 12\rangle^4}{\langle 13\rangle\langle 32\rangle\langle 24\rangle\langle 41\rangle} \tag{215}$$

The component $(22)(11)(22)(11)$, $(22)(11)(11)(22)$ and others are the same except for the helicity configuration.

The next important component is $(12)(21)(11)(22)$

$$(12)(21)(11)(22) \quad \Rightarrow \quad -2[31\rangle\langle 24\rangle[32]\langle 14\rangle \tag{216}$$

so that the partial amplitude equals to

$$\frac{-2[31\rangle\langle 24\rangle[32]\langle 14\rangle}{\langle 12\rangle[21]\langle 23\rangle[32]}=\frac{-2\,\langle 24\rangle\langle 14\rangle}{\langle 12\rangle\langle 23\rangle}\times\frac{[31]}{[21]} \tag{217}$$

Also from the momentum conservation, we know that

$$\frac{[31]}{[21]}=-\frac{\langle 42\rangle}{\langle 43\rangle} \tag{218}$$

then the partial amplitude equals to

$$\frac{2\langle 24\rangle^2\langle 14\rangle^2}{\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 41\rangle} \tag{219}$$

which is expected for 4D color ordered amplitude $A_4[1,2,3^+,4^-]$.

Then we need to consider a special kind of component – all scalar amplitude. The scalar component is (12) or (21), so we can list all possible components

$$(12)(12)(12)(12) \qquad (21)(21)(21)(21) \tag{220}$$
$$(12)(12)(21)(21) \qquad (21)(21)(12)(12) \tag{221}$$
$$(12)(21)(12)(21) \qquad (21)(12)(21)(12) \tag{222}$$
$$(12)(21)(21)(12) \qquad (21)(12)(12)(21) \tag{223}$$

In order to make components explicit, we first list the crossing term appearing in (202)

$$-2\begin{pmatrix}[21] & \\ & \langle 12\rangle\end{pmatrix}_{ab}\begin{pmatrix}[43] & \\ & \langle 34\rangle\end{pmatrix}_{cd}\begin{pmatrix}[31] & \\ & \langle 13\rangle\end{pmatrix}_{ac}\begin{pmatrix}[42] & \\ & \langle 24\rangle\end{pmatrix}_{bd} \tag{224}$$
$$-2\begin{pmatrix}[31] & \\ & \langle 13\rangle\end{pmatrix}_{ac}\begin{pmatrix}[42] & \\ & \langle 24\rangle\end{pmatrix}_{bd}\begin{pmatrix}[41] & \\ & \langle 14\rangle\end{pmatrix}_{ad}\begin{pmatrix}[32] & \\ & \langle 23\rangle\end{pmatrix}_{bc} \tag{225}$$
$$+2\begin{pmatrix}[21] & \\ & \langle 12\rangle\end{pmatrix}_{ab}\begin{pmatrix}[43] & \\ & \langle 34\rangle\end{pmatrix}_{cd}\begin{pmatrix}[41] & \\ & \langle 14\rangle\end{pmatrix}_{ad}\begin{pmatrix}[32] & \\ & \langle 23\rangle\end{pmatrix}_{bc}. \tag{226}$$

Then we can read off the first and second component

$$\begin{aligned}(12)(12)(12)(12) =& [21][43]\langle 12\rangle\langle 34\rangle + [31][42]\,\langle 13\rangle\langle 24\rangle + [41][32]\,\langle 14\rangle\langle 23\rangle \\ &- 2[21][43]\,\langle 13\rangle\langle 24\rangle - 2[31][42]\,\langle 14\rangle\langle 23\rangle + 2[21][43]\,\langle 14\rangle\langle 23\rangle\,.\end{aligned} \tag{227}$$

This is not so clear now so we need to write down all components

$$\begin{aligned}(21)(21)(21)(21) =& [21][43]\langle 12\rangle\langle 34\rangle + [31][42]\,\langle 13\rangle\langle 24\rangle + [41][32]\,\langle 14\rangle\langle 23\rangle \\ &- 2\,\langle 12\rangle\langle 34\rangle\,[31][42] - 2\,\langle 13\rangle\langle 24\rangle\,[41][32] + 2\,\langle 12\rangle\langle 34\rangle\,[41][32].\end{aligned} \tag{228}$$

and

$$(12)(12)(21)(21) = [21]\langle 34\rangle\langle 12\rangle[43] \qquad (21)(21)(12)(12) = \langle 12\rangle[43][21]\langle 34\rangle \tag{229}$$
$$(12)(21)(12)(21) = [31]\langle 24\rangle\langle 13\rangle[42] \qquad (21)(12)(21)(12) = \langle 13\rangle[42][31]\langle 24\rangle \tag{230}$$
$$(12)(21)(21)(12) = [41]\langle 23\rangle\langle 14\rangle[32] \qquad (21)(12)(12)(21) = \langle 14\rangle[32][41]\langle 23\rangle \tag{231}$$

Finally, we want to consider the component

$$(12)(12)(11)(11) \tag{232}$$

A Detailed explaination of special 5D spinor variables

Lemma : *Momentum consevation for 3-particle amplitude implies*

$$y^{12}{}_{ab} := 1^{A}{}_{a}2_{Ab} = y^{1}{}_{a}y^{2}_{b} = 1_a 2_b \tag{233}$$
$$y^{23}{}_{bc} := 2^{A}{}_{b}3_{Ac} = y^{2}{}_{b}y^{3}_{c} = 2_b 3_c \tag{234}$$
$$y^{31}{}_{ca} := 3^{A}{}_{c}1_{Aa} = y^{3}{}_{c}y^{1}_{a} = 3_c 1_a \tag{235}$$

Proof. First note that $y_{ab}^{(i)(i)} = 0$, e.g.

$$\Omega_{AB}1_a^A 2_b^B = C\epsilon_{ab}(\text{from antisymmetry of ab}) \quad \text{and} \quad C \propto \Omega_{AB}1_a^A 1_b^B \epsilon^{ab} \propto \Omega_{AB}p_1^{AB} = 0 \tag{236}$$

Another property of y is that the determinant of this y^{ij} vanishes such that

$$\mathrm{Det}(y^{12}) = \frac{1}{2}\epsilon^{ab}\epsilon^{cd}{y^{12}}_{ac}{y^{12}}_{bd} = \frac{1}{2}{y^{12}}_{ac}y^{12ac} = \frac{1}{2}p_1^{AB}p_{2AB} = 2(p_1 \cdot p_2) = 0 \tag{237}$$

Thus it is a rank 1 quantity, which can be decomposed into the outer product of 2-spinors:

$$\begin{aligned} {y^{12}}_{ab} &= {y^1}_a {y^2}_b && (238)\\ {y^{23}}_{bc} &= {\Upsilon^2}_b {\Upsilon^3}_c && (239)\\ {y^{31}}_{ca} &= {\Upsilon^3}_c {\Upsilon^1}_a && (240) \end{aligned}$$

Note that not all of there $\mathrm{SU}(2)$ spinors are independent, indeed it is only a 2-dimension vector space. By momentum conservation

$${y^{12}}_a{}^b {y^{23}}_b{}^c = {1^A}_a {2_A}^b {2^B}_b {3_B}^c = {1^A}_a {p_{aA}}^B {3_B}^c = {1^A}_a(-{p_{1A}}^B - {p_{3A}}^B){3_B}^c = 0 \tag{241}$$

Thus $\Upsilon^{2b}{y^2}_b = 0$,so ${\Upsilon^2}_b \propto {y^2}_b$ (note the metric we are using now is antisymmetric tensor ϵ_{ab}). Similarly,

$$\begin{aligned} {y^{23}}_b{}^c {y^{31}}_c{}^a = 0 &\Rightarrow {\Upsilon^3}_c \propto {y^3}_c && (242)\\ {y^{31}}_c{}^a {y^{12}}_a{}^b = 0 &\Rightarrow {\Upsilon^1}_a \propto {y^1}_a && (243) \end{aligned}$$

Therefore, up to some constants A,B,C, we have

$$\begin{aligned} {y^{12}}_{ab} &= {y^1}_a y_b^2 && (244)\\ {y^{23}}_{bc} &= B{y^2}_b y_c^3 && (245)\\ {y^{31}}_{ca} &= C{y^3}_c A y_a^1 && (246) \end{aligned}$$

As long as we are working over $\mathbb{C}$, we can always rescale them to obtain

$$\begin{aligned} {y^{12}}_{ab} &= {y^1}_a y_b^2 && (247)\\ {y^{23}}_{bc} &= {y^2}_b y_c^3 && (248)\\ {y^{31}}_{ca} &= {y^3}_c y_a^1 && (249) \end{aligned}$$

here each y are determined uniquely, up to an overall flip $(y^1, y^2, y^3) \mapsto (-y^1, -y^2, -y^3)$.

The space of 2-dimension representation of $\mathrm{SU}(2)$ is too small to ensure the hold of following equality

$$\delta_A \coloneqq 1^a 1_{Aa} = 2^b 2_{Ab} = 3^c 3_{Ac} \tag{250}$$

δ_A is same for all particles.

Proof. From the 3-particle momentum conservation

$$0 = 1_A{}^a 1_{Ba} + 2_A{}^b 2_{Bb} + 3_A{}^c 3_{Bc} \tag{251}$$

$$0 = \underbrace{1^A{}_a (1_A{}^{a'}}_{=0} 1_{Ba'} + 2_A{}^b 2_{Bb} + 3_A{}^c 3_{Bc}) \tag{252}$$

$$0 = (1_a 2^b) 2_{Bb} + (-1_a 3^c) 3_{Bc} \tag{253}$$

$$0 = 1_a (2^b 2_{Bb} - 3^c 3_{Bc}) \tag{254}$$

$$\Rightarrow 2^b 2_{Bb} = 3^c 3_{Bc} \tag{255}$$

Cyclically premutating the indices, we can obtain $1^a 1_{Ba} = 2^b 2_{Bb} = 3^c 3_{Bc}$.

A useful equality can be obtained

$$\delta_A \lambda_(i)^A{}_b \propto y^{ia} y^i{}_b = 0 \tag{256}$$

B χ in 5D

Since $\delta_A = 1^a 1_{Aa} = 2^b 2_{Ab} = 3^c 3_{Ac}$, we can evaluate χ_3 by

$$p_2{}^{AB} 3_B{}^c = 2^{Ab} 2^B{}_b 3_B{}^c = 2^{Ab} 2_b 3^c = -\delta^A 3^c = 3^{Ac'} 3_{c'} 3^c = \underbrace{(3^c 3_{c'})}_{\chi_3} 3^{Ac'} \tag{257}$$

Cycling the particle labels, we can obtain the χ for each particle

$$p_1{}^{AB} 2_B{}^b = (2^b 2_{b'}) 2^{Ab'} \tag{258}$$

$$p_3{}^{AB} 1_B{}^a = (1^a 1_{a'}) 1^{Aa'} \tag{259}$$

Other combinations can be simply obtained by momentum conservation,like $p_1{}^{AB} 3_B{}^c = -p_2{}^{AB} 3_B{}^c = -\chi^c_{3c'} 3^{Ac'}$.